AHEAD IN
ANIMAL HEALTH

API Documentation

Retrieve Lab Results DECIDE

Data Exchange with Royal GD

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Source: Royal GD
Version: 1.0
Status: Final

Contents

Version History

Date	Version	Change	Author
28-11-2024	0.1	Initial version.	Eelco Tulp
03-12-2024	0.2	Review of Manon Holstege, Leyla Yanbeyi and Folkert Algra processed.	Eelco Tulp
04-12-2024	0.3	Use the POST method to retrieve the token. Use the GET method when calling the service's endpoints.	Eelco Tulp
11-02-2026	1.0	Final version.	Eelco Tulp

1 General

1.1 Introduction

Before you lies the technical description of the implementation of the web service, Retrieve Lab Results for DECIDE from Royal GD (from now on referred to as GD). This allows you to retrieve lab results using an API from GD for the benefit of DECIDE.

This document does not describe:

- The service delivery must meet the service levels stated in the document SLA Use of Web services (APIs) from Royal GD. This document also describes the process for connecting to a web service from GD.
- Contract information, such as terms, validity, price agreements, etc., is stated in the associated data delivery agreements.

1.2 Reading Guide

Chapter 2 contains a glossary of terms and abbreviations; Chapter 3 covers the general technical specifications. Finally, Chapter 4 includes the definition of the message.

2 Glossary

Begrip	Omschrijving
Base64	Base64 is a way to convert binary code to ASCII characters.
HTTP	Hypertext Transfer Protocol (HTTP) is het protocol voor de communicatie tussen een webclient (meestal een webbrowser of een app) en een webserver.
JSON	Hypertext Transfer Protocol (HTTP) is the protocol for communication between a web client (usually a web browser or an app) and a web server.
JWT	JSON Web Token.
OAuth	OAuth (Open Authorization) is an open standard for authorisation.
TLS	Transport Layer Security and its predecessor, Secure Sockets Layer, are encryption protocols protecting computer communications.
URL	A Uniform Resource Locator (URL for short) is a structured name that refers to a piece of data.

3 Technical API Specification

3.1 Operation of the Web Service

This web service offers the possibility of requesting lab results from GD via an API for DECIDE.

3.2 Authentication/Authorisation

GD's web services are secured with the OAuth 2.0 protocol. All traffic to and from the service is via TLS.

The production environment URL of the token service is
https://services.gdanimalhealth.com/oauth2/resources/tokenservice

The test environment URL of the token service is:
https://services-acc.gdanimalhealth.com/oauth2/resources/tokenservice

NB:
Use the POST method to retrieve the token.

The username/password supplied by GD must be included as basic authentication in the HTTP header in the request message to the token service.
Curl is a tool for making HTTP requests from the command line. Below is an example of a call to the token service using curl.

```
curl --location
'https://services.gdanimalhealth.com/oauth2/resources/tokenservice' \
--header 'Content-Type: application/x-www-form-urlencoded' \
--header 'Authorization: Basic <base-64-encoded-username-password>' \
--data-urlencode 'grant_type=client_credentials'
```

Below is an example of calling our web services using the previously retrieved token.

```
curl --location 'https://services.gdanimalhealth.com/<endpoint_service>' \
--header 'Authorization: Bearer <access_token>' \
--header 'Content-Type: application/json; charset=utf-8' \
```

The JSON response will contain an 'access_token' field. The value of this field must be sent with the requests.

JWT bearer tokens are used for this. The service must be called with OAuth 2.0 by adding the HTTP header 'Authorization' to the request with the value 'Bearer <access_token>' (<access_token> here as a placeholder for the previously retrieved access_token).

3.3 Environments

To use GD's test and production environment, you must first be connected (see SLA Use web services (APIs) of Royal GD).

You must use the GET method when calling the service's endpoints in one of the environments below.

3.3.1 Test Environment

For testing purposes, we have a test environment. The URL for this is:

https://services-acc.gdanimalhealth.com/api/Decide/1.0/labresults

3.3.2 Production Environment

The production environment URL is:

https://services.gdanimalhealth.com/api/Decide/1.0/labresults

3.4 Error Handling

Errors are returned with an HTTP status of 4XX or 5XX (depending on the type of error). The web service will (if possible) compose the ServiceStatus object in the JSON response with a code and description of the error.

For example, if the JSON request does not meet the specification, it can lead to an HTTP-400 (Bad Request) error code and accompanying text in the JSON response.

3.5 Reliability of the Web Service

The reliability of the web service is guaranteed in several ways:

- Identification and authentication of the reporter: After authentication, the web service will authorise the requesting party by retrieving a token and relationship identification provided in the message.
- Security of message traffic: To optimally secure the traffic between your application and the web service, so-called HTTPS and TLS connections are used. To establish an HTTPS or TLS connection, the internet settings of the machine that establishes contact with the web service must be set so that this machine accepts a certificate.

4 Message Specifications

The message shown in this chapter is a functional representation of the actual JSON message. A sample JSON message will be provided separately from this message book.

4.1 Request- and Response Messages

4.1.1 Class Diagram

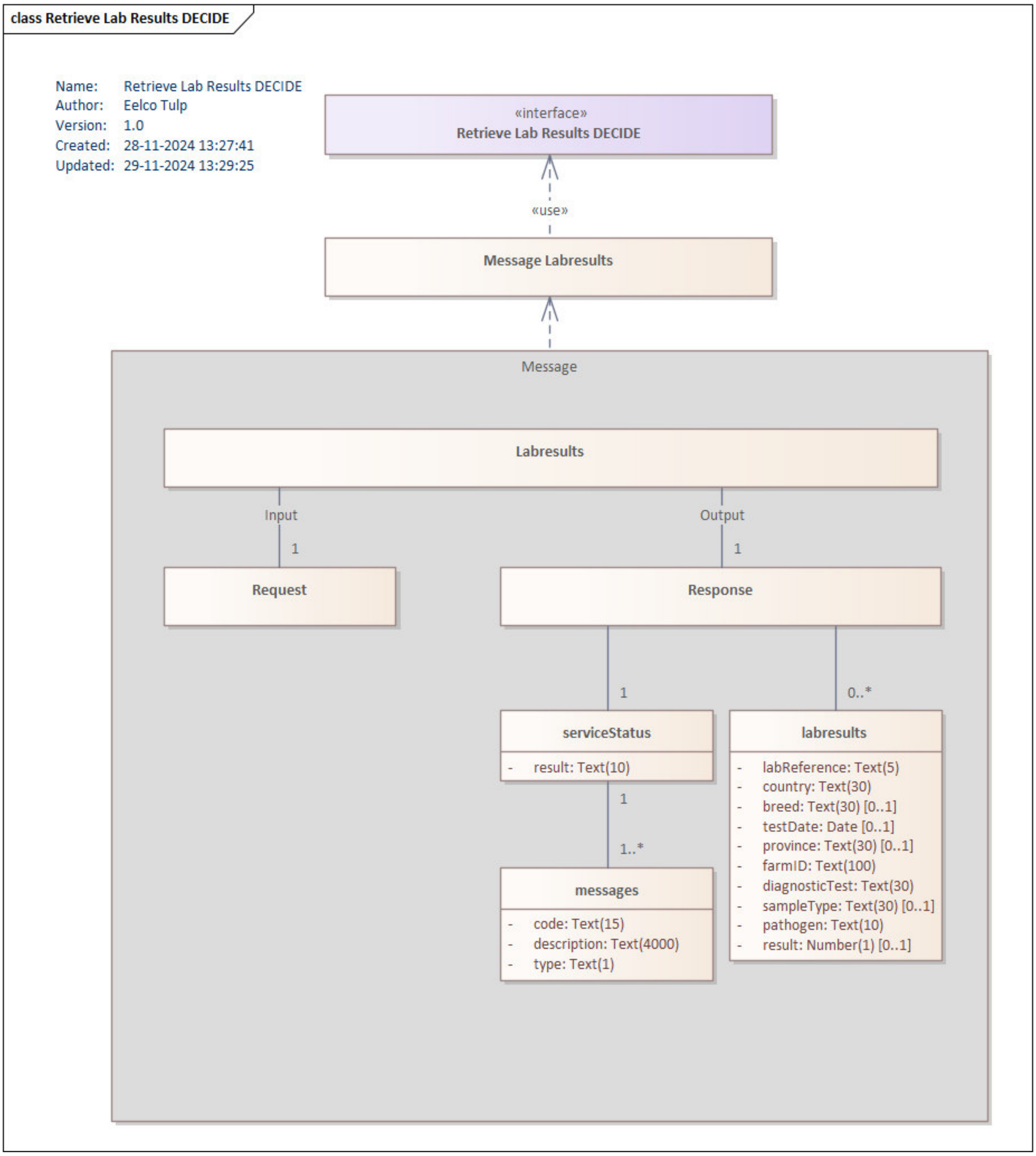

Diagram explanation:
This class diagram shows the structure of the message that can be used to request lab results from GD for DECIDE.

4.1.2 Input
The web service has no input. When called, the most recent data is constantly retrieved and returned.

4.1.3 Message Definition
Below are the various definitions of the message content.

***Class* serviceStatus**
Status information of the response of an service operation.

Explanation:
Concerns the result of the service processing (OK or NOT OK) and any error messages that occurred during the execution of the service operation.

Attribuut		
Naam	**Beschrijving**	**Datatype**
result	The result of the service operation. Possible values: • OK • NOT OK	Text(10) [1..1]

***Class* messages**
Message(s) returned from service processing.

Attribuut		
Naam	**Beschrijving**	**Datatype**
code	The code of message Explanation: For example: "SRV-00000".	Text(15) [1..1]
description	The body of the message Explanation: For example: "Success.".	Text(4000) [1..1]
type	Type of the message Explanation: Possible values: • I = Information • E = Error	Text(1) [1..1]

Class labresults The lab results that are prepared for DECIDE. The message will include a line for each lab result. See the class diagram and the associated definitions for the elements and attributes.		
Attribuut		
Naam	**Beschrijving**	**Datatype**
labReference	Reference to the performing laboratory.	Text(5) [1..1]
country	Country of origin of the sample.	Text(30) [1..1]
breed	Production type. Position in the production chain.	Text(30) [0..1]
testDate	The date the test was authorised.	Date [0..1]
province	Province within the country of origin where the sample was taken.	Text(30) [0..1]
farmID	The encrypted identification of the farm from which the sample originated.	Text(100) [1..1]
diagnosticTest	The test performed on the sample to which the lab result relates.	Text(30) [1..1]
sampleType	The sample material type.	Text(30) [0..1]
pathogen	The pathogen type.	Text(10) [1..1]
result	The result of the laboratory research.	Number(1) [0..1]

4.2 JSON Message Sample

A sample JSON message will be provided separately from this API documentation.